

CO4 -AT2 - Coding Golf Contest

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QUESTIONS

- Q1 Quick Sort 25 marks
- Q2 Bubble Sort 25 marks
- Q3 Insertion Sort 25 marks
- Q4 Merge Sort 25 marks

4/4 answered

Q1 Quick Sort

25 marks

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

Your Code

```
1 #include <stdio.h>
2 void swap(int* a, int* b) {
3     int t = *a;
4     *a = *b;
5     *b = t;
6 }
7 int partition(int arr[], int low, int high) {
8     int pivot = arr[high];
9     int i = (low - 1);
10
11     for (int j = low; j <= high - 1; j++) {
12         if (arr[j] < pivot) {
13             i++;
14             swap(&arr[i], &arr[j]);
15         }
16     }
17     swap(&arr[i + 1], &arr[high]);
18     return (i + 1);
19 }
20 void quickSort(int arr[], int low, int high) {
21     if (low < high) {
22         int pi = partition(arr, low, high);
23         quickSort(arr, low, pi - 1);
24         quickSort(arr, pi + 1, high);
25     }
26 }
27 int main() {
```

Input

stdin input

Output

9 18 23 32 50 61

- Q2 Bubble Sort 25 marks
- Q3 Insertion Sort 25 marks
- Q4 Merge Sort 25 marks

4/4 answered

Your Code

```
1 #include <stdio.h>
2 void swap(int* a, int* b) {
3     int t = *a;
4     *a = *b;
5     *b = t;
6 }
7 int partition(int arr[], int low, int high) {
8     int pivot = arr[high];
9     int i = (low - 1);
10
11     for (int j = low; j <= high - 1; j++) {
12         if (arr[j] < pivot) {
13             i++;
14             swap(&arr[i], &arr[j]);
15         }
16     }
17     swap(&arr[i + 1], &arr[high]);
18     return (i + 1);
19 }
20 void quickSort(int arr[], int low, int high) {
21     if (low < high) {
22         int pi = partition(arr, low, high);
23         quickSort(arr, low, pi - 1);
24         quickSort(arr, pi + 1, high);
25     }
26 }
27 int main() {
28     int arr[] = {50, 23, 8, 18, 61, 32};
29     int n = sizeof(arr) / sizeof(arr[0]);
30     quickSort(arr, 0, n - 1);
31     for (int i = 0; i < n; i++) {
32         printf("%d ", arr[i]);
33     }
34     printf("\n");
35     return 0;
36 }
```

Input

stdin input

Output

8 18 23 32 50 61

QUESTIONS

- Q1**
Quick Sort
25 marks
- Q2**
Bubble Sort
25 marks
- Q3**
Insertion Sort
25 marks
- Q4**
Merge Sort
25 marks

4/4 answered

Q2 Bubble Sort

25 marks

Write a program that implements the bubble sort.

Your Code

```
1 #include <stdio.h>
2
3 void bubbleSort(int arr[], int n) {
4     for (int i = 0; i < n - 1; i++) {
5         int swapped = 0;
6         for (int j = 0; j < n - i - 1; j++) {
7             if (arr[j] > arr[j + 1]) {
8                 int temp = arr[j];
9                 arr[j] = arr[j + 1];
10                arr[j + 1] = temp;
11                swapped = 1;
12            }
13        }
14        if (!swapped) break;
15    }
16 }
17
18 int main() {
19     int n;
20     scanf("%d", &n);
21     int arr[n];
22     for (int i = 0; i < n; i++) {
23         scanf("%d", &arr[i]);
24     }
25     bubbleSort(arr, n);
26     for (int i = 0; i < n; i++) {
27         printf("%d ", arr[i]);
28     }
29     return 0;
30 }
```

Input

5
2 3 1 4 5

Output

1 2 3 4 5

QUESTIONS

- Q1**
Quick Sort
25 marks
- Q2**
Bubble Sorting
25 marks
- Q3**
Insertion Sort
25 marks
- Q4**
Merge Sort
25 marks

4/4 answered

Q3 Insertion Sort

25 marks

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

Your Code

```
1 #include <stdio.h>
2 void insertionSort(int arr[], int n) {
3     int i, key, j;
4     for (i = 1; i < n; i++) {
5         key = arr[i];
6         j = i - 1;
7         while (j >= 0 && arr[j] > key) {
8             arr[j + 1] = arr[j];
9             j = j - 1;
10        }
11        arr[j + 1] = key;
12    }
13 }
14 void printArray(int arr[], int n) {
15     for (int i = 0; i < n; i++) {
16         printf("%d ", arr[i]);
17     }
18     printf("\n");
19 }
20 int main() {
21     int arr[] = {9, 5, 1, 4, 3};
22     int n = sizeof(arr) / sizeof(arr[0]);
23     insertionSort(arr, n);
24     printArray(arr, n);
25     return 0;
26 }
```

Submitted

Input

stdin input

Output

1 3 4 5 9

QUESTIONS

- Q1 Quick Sort 25 marks
- Q2 Bubble Sorting 25 marks
- Q3 Insertion Sort 25 marks
- Q4 Merger Sort 25 marks

4/4 answered

Q4 Merger Sort

25 marks

Apply Merge Sort to the array 38, 27, 43, 3, 9, 62, 10.

Your Code

```
1 #include <stdio.h>
2 void merge(int arr[], int l, int m, int r) {
3     int i, j, k;
4     int n1 = m - l + 1;
5     int n2 = r - m;
6     int L[n1], R[n2];
7     for (i = 0; i < n1; i++)
8         L[i] = arr[l + i];
9     for (j = 0; j < n2; j++)
10        R[j] = arr[m + 1 + j];
11     i = 0;
12     j = 0;
13     k = l;
14     while (i < n1 && j < n2) {
15         if (L[i] <= R[j]) {
16             arr[k] = L[i];
17             i++;
18         } else {
19             arr[k] = R[j];
20             j++;
21         }
22         k++;
23     }
24     while (i < n1) {
25         arr[k] = L[i];
26         i++;
27         k++;
28     }
29     while (j < n2) {
30         arr[k] = R[j];
```

Input

stdin input

Output

3 9 10 27 38 43 62

```
16         arr[k] = L[i];
17         i++;
18     } else {
19         arr[k] = R[j];
20         j++;
21     }
22     k++;
23 }
24 while (i < n1) {
25     arr[k] = L[i];
26     i++;
27     k++;
28 }
29 while (j < n2) {
30     arr[k] = R[j];
31     j++;
32     k++;
33 }
34 }
35 void mergeSort(int arr[], int l, int r) {
36     if (l < r) {
37         int m = l + (r - l) / 2;
38         mergeSort(arr, l, m);
39         mergeSort(arr, m + 1, r);
40         merge(arr, l, m, r);
41     }
42 }
43 int main() {
44     int arr[] = {38, 27, 43, 3, 9, 82, 10};
45     int arr_size = sizeof(arr) / sizeof(arr[0]);
46     mergeSort(arr, 0, arr_size - 1);
47     for (int i = 0; i < arr_size; i++) {
48         printf("%d ", arr[i]);
49     }
50     printf("\n");
51     return 0;
52 }
```

Submitted